

Poisson Summation

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1 Poisson Summation

Let Λ be a complete integral lattice in $\mathbb{R}^n$ with fundamental domain $\mathcal{M}$ and denote the dual integral lattice by $\Lambda^\vee$. Recall that the $\lambda^\vee$ -th Fourier coefficient $\hat{f}(\lambda^\vee)$ of $f(\mathbf{x})$ is

$$\hat{f}(\lambda^\vee) = \int_{\mathcal{M}} f(\mathbf{x}) e^{-2\pi i \langle \lambda^\vee, \mathbf{x} \rangle} d\mathbf{x},$$

while the Fourier series $S_f(x)$ is

$$S_f(\mathbf{x}) = \frac{1}{V_\Lambda} \sum_{\lambda^\vee \in \Lambda^\vee} \hat{f}(\lambda^\vee) e^{2\pi i \langle \lambda^\vee, \mathbf{x} \rangle}.$$

There are two natural methods of building a function from $f(\mathbf{x})$ that is λ -periodic. The first is to average $f(\mathbf{x})$ over all translates by elements of Λ while the second is to consider the Fourier series of $f(\mathbf{x})$. This gives us the two series

$$\sum_{\lambda \in \Lambda} f(\mathbf{x} + \lambda) \quad \text{and} \quad \frac{1}{V_\Lambda} \sum_{\lambda^\vee \in \Lambda^\vee} (\mathcal{F}f)(\lambda^\vee) e^{2\pi i \langle \lambda^\vee, \mathbf{x} \rangle}.$$

The link between the Fourier transform and Fourier series is given by **Poisson summation** which says that these two expressions are the same under some mild assumptions.

Theorem (Poisson summation). *Suppose Λ is a complete integral lattice in $\mathbb{R}^n$, $f(\mathbf{x})$ is absolutely integrable on $\mathbb{R}^n$, and the function*

$$F(\mathbf{x}) = \sum_{\lambda \in \Lambda} f(\mathbf{x} + \lambda),$$

is smooth. Then

$$\sum_{\lambda \in \Lambda} f(\mathbf{x} + \lambda) = \frac{1}{V_\Lambda} \sum_{\lambda^\vee \in \Lambda^\vee} (\mathcal{F}f)(\lambda^\vee) e^{2\pi i \langle \lambda^\vee, \mathbf{x} \rangle},$$

and

$$\sum_{\lambda \in \Lambda} f(\lambda) = \frac{1}{V_\Lambda} \sum_{\lambda^\vee \in \Lambda^\vee} (\mathcal{F}f)(\lambda^\vee).$$

Proof. Fix a basis $\lambda_1, \dots, \lambda_n$ for Λ and let P be the associated generator matrix. Then $\Lambda = P\mathbb{Z}^n$ and $\Lambda^\vee = (P^{-1})^t \mathbb{Z}^n$. Letting $F_P(\mathbf{x}) = F(P\mathbf{x})$, we may write

$$F_P(\mathbf{x}) = \sum_{\mathbf{n} \in \mathbb{Z}^n} f(P\mathbf{x} + P\mathbf{n}).$$

$F_P(\mathbf{x})$ is smooth and 1-periodic in each component because $F(\mathbf{x})$ is smooth and invariant under translation by Λ . Whence $F_P(\mathbf{x})$ admits a Fourier series converging uniformly everywhere to $F_P(\mathbf{x})$. Moreover, $F_P(\mathbf{x})$ is absolutely integrable on $[0, 1]^n$. We will compute the $\mathbf{t}$ -th Fourier coefficient of $F_P(\mathbf{x})$ given by

$$\hat{F}_P(\mathbf{t}) = \int_{[0,1]^n} F_P(\mathbf{x}) e^{-2\pi i \langle \mathbf{t}, \mathbf{x} \rangle} d\mathbf{x}$$

The absolute integrability of $f(\mathbf{x})$ permits the interchange of sum and integral to obtain

$$\sum_{\mathbf{n} \in \mathbb{Z}^n} \int_{[0,1]^n} f(P\mathbf{x} + P\mathbf{n}) e^{-2\pi i \langle \mathbf{t}, \mathbf{x} \rangle} d\mathbf{x}.$$

The change of variables $\mathbf{x} \mapsto P^{-1}\mathbf{x}$ gives

$$\frac{1}{V_\Lambda} \sum_{\mathbf{n} \in \mathbb{Z}^n} \int_{P[0,1]^n} f(\mathbf{x} + P\mathbf{n}) e^{-2\pi i \langle (P^{-1})^t \mathbf{t}, \mathbf{x} \rangle},$$

because the Jacobian matrix is P^{-1} with Jacobian determinant $\frac{1}{V_\Lambda}$. This expression is simply

$$\frac{1}{V_\Lambda} \int_{\mathbb{R}^n} f(\mathbf{x}) e^{-2\pi i \langle (P^{-1})^t \mathbf{t}, \mathbf{x} \rangle} d\mathbf{x} = \frac{1}{V_\Lambda} (\mathcal{F}f)((P^{-1})^t \mathbf{t}).$$

Whence

$$\sum_{\lambda \in \Lambda} f(P\mathbf{x} + \lambda) = \frac{1}{V_\Lambda} \sum_{\mathbf{t} \in \mathbb{Z}^n} (\mathcal{F}f)((P^{-1})^t \mathbf{t}) e^{2\pi i \langle \mathbf{t}, \mathbf{x} \rangle}.$$

Changing variables $\mathbf{x} \mapsto P^{-1}\mathbf{x}$ results in

$$\sum_{\lambda \in \Lambda} f(\mathbf{x} + \lambda) = \frac{1}{V_\Lambda} \sum_{\lambda^\vee \in \Lambda^\vee} (\mathcal{F}f)(\lambda^\vee) e^{2\pi i \langle \lambda^\vee, \mathbf{x} \rangle}.$$

This proves the first statement. Setting $\mathbf{x} = \mathbf{0}$ proves the second statement. $\square$

For convenience, we state Poisson summation in the simplified case for the complete integral lattice $\mathbb{Z}^n$ as a corollary since it is how Poisson summation is usually applied.

Corollary 1.1. *Suppose $f(\mathbf{x})$ is absolutely integrable on $\mathbb{R}^n$, and the function*

$$F(\mathbf{x}) = \sum_{\mathbf{n} \in \mathbb{Z}^n} f(\mathbf{x} + \mathbf{n}),$$

is smooth. Then

$$\sum_{\mathbf{n} \in \mathbb{Z}^n} f(\mathbf{x} + \mathbf{n}) = \sum_{\mathbf{t} \in \mathbb{Z}^n} (\mathcal{F}f)(\mathbf{t}) e^{2\pi i \langle \mathbf{t}, \mathbf{x} \rangle},$$

and

$$\sum_{\mathbf{n} \in \mathbb{Z}^n} f(\mathbf{n}) = \sum_{\mathbf{t} \in \mathbb{Z}^n} (\mathcal{F}f)(\mathbf{t}).$$

Proof. This is Poisson summation for the complete integral lattice $\mathbb{Z}^n$ since $\mathbb{Z}^n$ is self-dual and $V_{\mathbb{Z}^n} = 1$. $\square$

Using the Dirichlet-Jordan test, we can prove a slightly stronger form of Poisson summation in the single variable setting for the complete integral lattice $\mathbb{Z}$.

Theorem 1.2. *Suppose $f(x)$ is absolutely integrable on $\mathbb{R}$, and the function*

$$F(x) = \sum_{n \in \mathbb{Z}}^* f(x + n),$$

satisfies the Dirichlet-Jordan test, where the $$ in the sum indicates that $f(x + n)$ is meant to represent the average of the left-hand and right-hand limits at jump discontinuities. Then*

$$\sum_{n \in \mathbb{Z}}^* f(x + n) = \sum_{t \in \mathbb{Z}} (\mathcal{F}f)(t) e^{2\pi i t x},$$

and

$$\sum_{n \in \mathbb{Z}}^* f(n) = \sum_{t \in \mathbb{Z}} (\mathcal{F}f)(t).$$

Proof. Observe that $F(x)$ is 1-periodic. As $F(x)$ satisfies the Dirichlet-Jordan test by assumption, it admits a Fourier series converging locally uniformly to $F(x)$ wherever $F(x)$ is continuous. In fact, by the construction of $F(x)$ and that the Fourier series of $F(x)$ converges to the average of the left-hand and right-hand limits at jump discontinuities, the Fourier series of $F(x)$ converges locally uniformly to $F(x)$ everywhere. Moreover, $F(x)$ is absolutely integrable on $[0, 1]$. We compute the t -th Fourier coefficient of $F(x)$ given by

$$\hat{F}(t) = \int_0^1 F(x) e^{-2\pi i t x} dx.$$

The absolute integrability of $f(x)$ permits the interchange of sum and integral to obtain

$$\int_{-\infty}^{\infty} f(x) e^{-2\pi i t x} dx = (\mathcal{F}f)(t).$$

Whence

$$\sum_{n \in \mathbb{Z}}^* f(x + n) = \sum_{t \in \mathbb{Z}} (\mathcal{F}f)(t) e^{2\pi i t x}.$$

This proves the first statement. Setting $x = 0$ proves the second statement. $\square$

In practical settings, we need a class of functions $f(\mathbf{x})$ for which the assumptions of Poisson summation hold. We say that $f(\mathbf{x})$ is of **Schwarz class** if $f \in C^\infty(\mathbb{R}^n)$ and $f(\mathbf{x})$ along with all of its partial derivatives have rapid decay. If $f(\mathbf{x})$ is of Schwarz class, the rapid decay implies that $f(\mathbf{x})$ and all of its derivatives are absolutely integrable over $\mathbb{R}^n$. Moreover, $F(\mathbf{x}) = \sum_{\mathbf{n} \in \mathbb{Z}^n} f(\mathbf{x} + \mathbf{n})$ and all of its derivatives are locally absolutely uniformly convergent by the Weierstrass M -test. The uniform limit theorem then ensures $F(\mathbf{x})$ is smooth and thus the conditions of Poisson summation are satisfied. So in most practical applications only only needed to ensure that $f(\mathbf{x})$ is Schwarz class.

We now introduce some explicit Schwarz class functions and compute their Fourier transforms. The classic example is $e^{-2\pi x^2}$. This function is particularly important because it is essentially its own Fourier transform.

Proposition 1.3. *Let $\alpha > 0$ and set $f(x) = e^{-2\pi\alpha x^2}$. Then*

$$(\mathcal{F}f)(\zeta) = \frac{e^{-\frac{\pi\zeta^2}{2\alpha}}}{\sqrt{2\alpha}}.$$

In particular, $e^{-\pi x^2}$ is its own Fourier transform.

Proof. Consider

$$(\mathcal{F}f)(\zeta) = \int_{-\infty}^{\infty} e^{-2\pi(\alpha x^2 + i\zeta x)} dx.$$

The change of variables $x \mapsto \frac{x}{\sqrt{\alpha}}$ transforms this integral into

$$\frac{1}{\sqrt{\alpha}} \int_{-\infty}^{\infty} e^{-2\pi\left(x^2 + \frac{i\zeta x}{\sqrt{\alpha}}\right)} dx.$$

Complete the square in the exponent by observing

$$x^2 + \frac{i\zeta x}{\sqrt{\alpha}} = \left(x + \frac{i\zeta}{2\sqrt{\alpha}}\right)^2 + \frac{\zeta^2}{4\alpha},$$

so that the previous integral is equal to

$$\frac{e^{-\frac{\pi\zeta^2}{2\alpha}}}{\sqrt{\alpha}} \int_{-\infty}^{\infty} e^{-2\pi\left(x + \frac{i\zeta}{2\sqrt{\alpha}}\right)^2} dx.$$

By rapid decay of the integrand, the change of variables $x \mapsto \frac{x}{\sqrt{2}} - \frac{i\zeta}{\sqrt{\alpha}}$ is permitted giving

$$\frac{e^{-\frac{\pi\zeta^2}{2\alpha}}}{\sqrt{2\alpha}} \int_{-\infty}^{\infty} e^{-\pi x^2} dx = \frac{e^{-\frac{\pi\zeta^2}{2\alpha}}}{\sqrt{2\alpha}},$$

because the remaining integral is the Gaussian integral. This proves the first statement. The second statement follows by taking $\alpha = \frac{1}{2}$. $\square$

The analog of $e^{-2\pi x^2}$ on $\mathbb{R}^n$ is $e^{-2\pi\langle \mathbf{x}, \mathbf{x} \rangle}$ which is Schwarz class because $e^{-2\pi x^2}$ is. We also obtain a generalization of the previous result this Schwarz class function as a corollary.

Corollary 1.4. *Let $\alpha > 0$ and set $f(\mathbf{x}) = e^{-2\pi\alpha\langle \mathbf{x}, \mathbf{x} \rangle}$. Then*

$$(\mathcal{F}f)(\boldsymbol{\xi}) = \frac{e^{-\frac{\pi\langle \boldsymbol{\xi}, \boldsymbol{\xi} \rangle}{2\alpha}}}{(2\alpha)^{\frac{n}{2}}}.$$

In particular, $e^{-\pi\langle \mathbf{x}, \mathbf{x} \rangle}$ is its own Fourier transform.

Proof. Applying Proposition 1.3 to each variable separately proves the first statement. The second statement follows upon setting $\alpha = \frac{1}{2}$. $\square$